Revision History

Version	Date	By	Change Dsecription
V1.0	2025-04-10	zuozhongkai	1:Initial release version

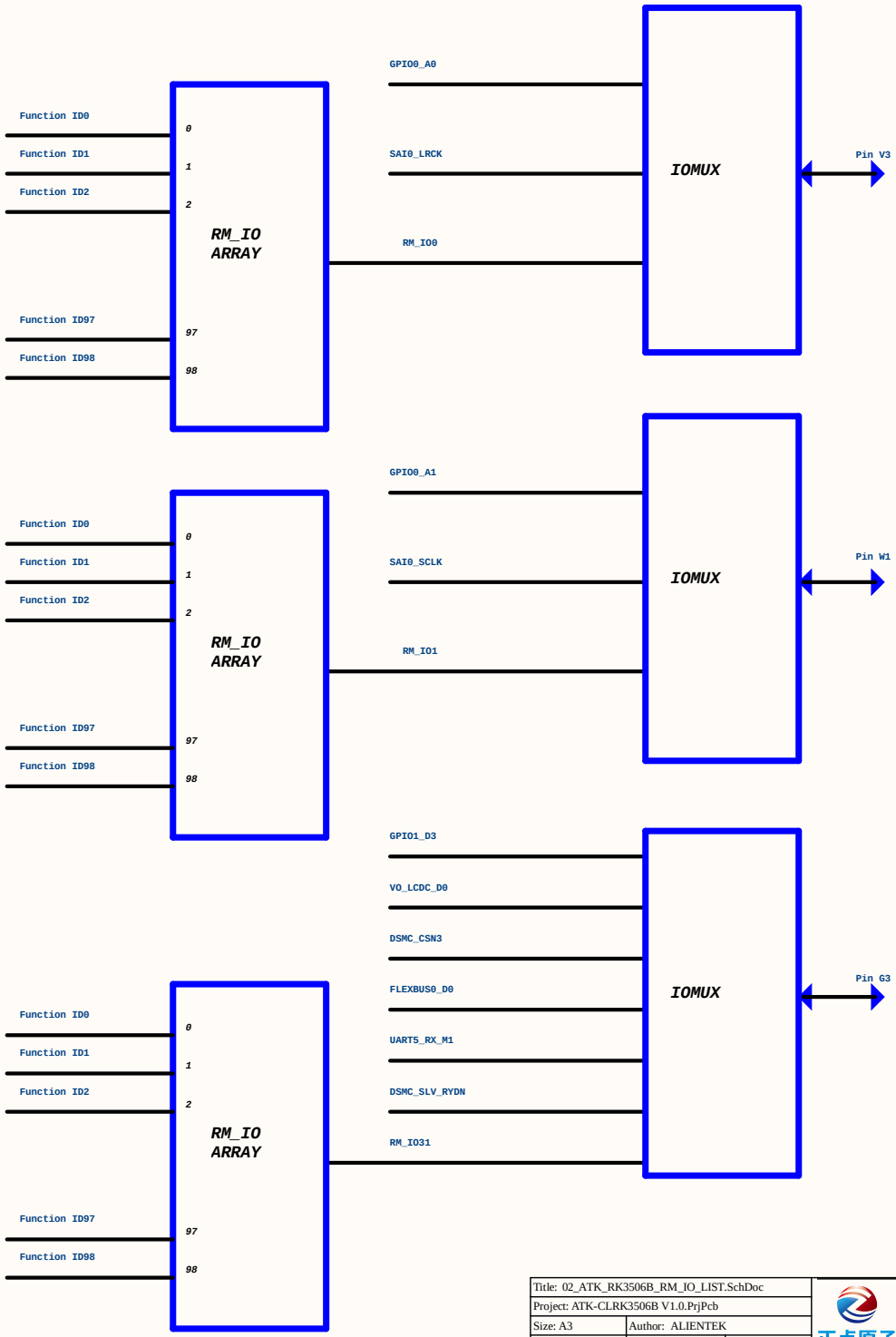
Title: 01_ATK_RK3506B_ReviSion_History.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 1 of 1

正点原子

RM_IO List of optional function ID0~98

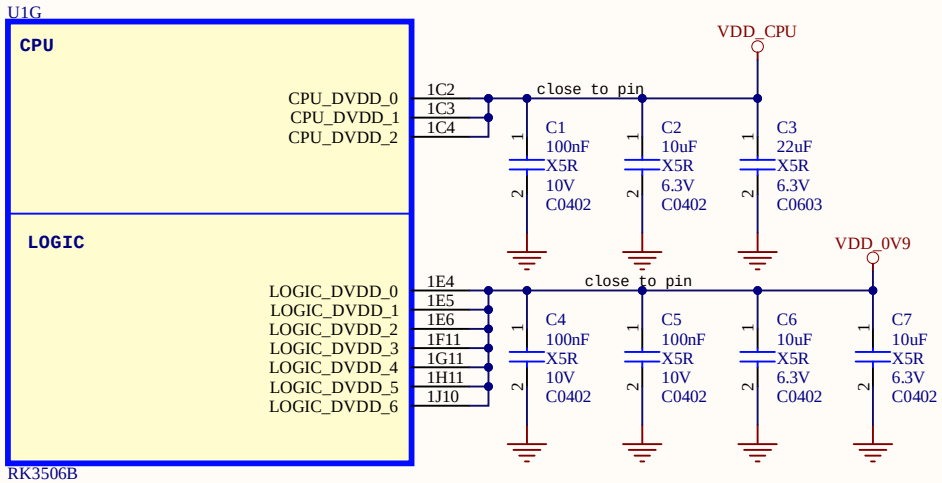
ID	function	ID	function	ID	function	ID	function
0	NC	26	CAN1_TX	51	SAI0_MCLK	77	TSADC_CTRL
1	UART1_TX	27	CAN1_RX	52	SAI0_SCLK	78	PWR_CTRL0(PMU_SLEEP)
2	UART1_RX	28	CAN0_TX	53	SAI0_LRCK	79	PWR_CTRL1(CORE_POWER_OFF)
3	UART2_TX	29	CAN0_RX	54	SAI0_SDI0	80	SPDIF_TX
4	UART2_RX	30	PWM0_CH0	55	SAI0_SDI1	81	SPDIF_RX
5	UART3_TX	31	PWM0_CH1	56	SAI0_SDI2	82	BIPCNTR_A0
6	UART3_RX	32	PWM0_CH2	57	SAI0_SDI3	83	BIPCNTR_A1
7	UART3_CTSN	33	PWM0_CH3	58	SAI0_SDO	84	BIPCNTR_A2
8	UART3_RTSN	34	PWM1_CH0	59	SAI1_MCLK	85	BIPCNTR_A3
9	UART4_TX	35	PWM1_CH1	60	SAI1_SCLK	86	BIPCNTR_A4
10	UART4_RX	36	PWM1_CH2	61	SAI1_LRCK	87	BIPCNTR_A5
11	UART4_CTSN	37	PWM1_CH3	62	SAI1_SDI	88	BIPCNTR_B0
12	UART4_RTSN	38	PWM1_CH4	63	SAI1_SDO0	89	BIPCNTR_B1
13	MIPI_DPHY_DSI_TX_TE	39	PWM1_CH5	64	SAI1_SDO1	90	BIPCNTR_B2
14	CLK_32K	40	PWM1_CH6	65	SAI1_SDO2	91	BIPCNTR_B3
15	I2C0_SCL	41	PWM1_CH7	66	SAI1_SDO3	92	BIPCNTR_B4
16	I2C0_SDA	42	TOUCH_KEYDRIVE	67	SPI0_CLK	93	BIPCNTR_B5
17	I2C1_SCL	43	TOUCH_KEY0	68	SPI0_MOSI	94	PDM_CLK1
18	I2C1_SDA	44	TOUCH_KEY1	69	SPI0_MISO	95	ETH_RMII0_PPSCLK
19	I2C2_SCL	45	TOUCH_KEY2	70	SPI0_CSN0	96	ETH_RMII0_PPSTRIG
20	I2C2_SDA	46	TOUCH_KEY3	71	SPI0_CSN1	97	ETH_RMII1_PPSCLK
21	PDM_CLK0	47	TOUCH_KEY4	72	SPI1_CLK	98	ETH_RMII1_PPSTRIG
22	PDM_SDI0	48	TOUCH_KEY5	73	SPI1_MOSI		
23	PDM_SDI1	49	TOUCH_KEY6	74	SPI1_MISO		
24	PDM_SDI2	50	TOUCH_KEY7	75	SPI1_CSN0		
25	PDM_SDI3			76	SPI1_CSN1		

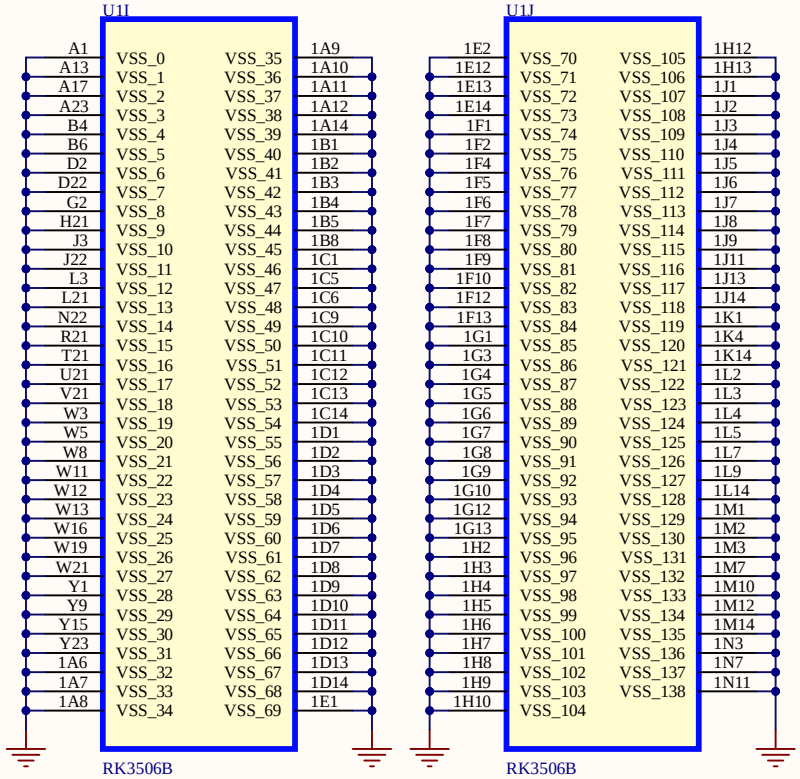
NOTE:
(1)RM_IOX(X=031) can select functions from function ID0ID98;
(2)RM_IOX(X=031) cannot repeatedly select the function ID0ID98;
(3)SAI、PDM、SPI signals should not cross the PMUIO and VCCIO domains;
(4)The signal of the same function in RM_IO is used across PMUIO and VCCIO domains, and the operating level needs to be consistent



Title: 02_ATK_RK3506B_RM_IO_LIST.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A3	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 2 of 14

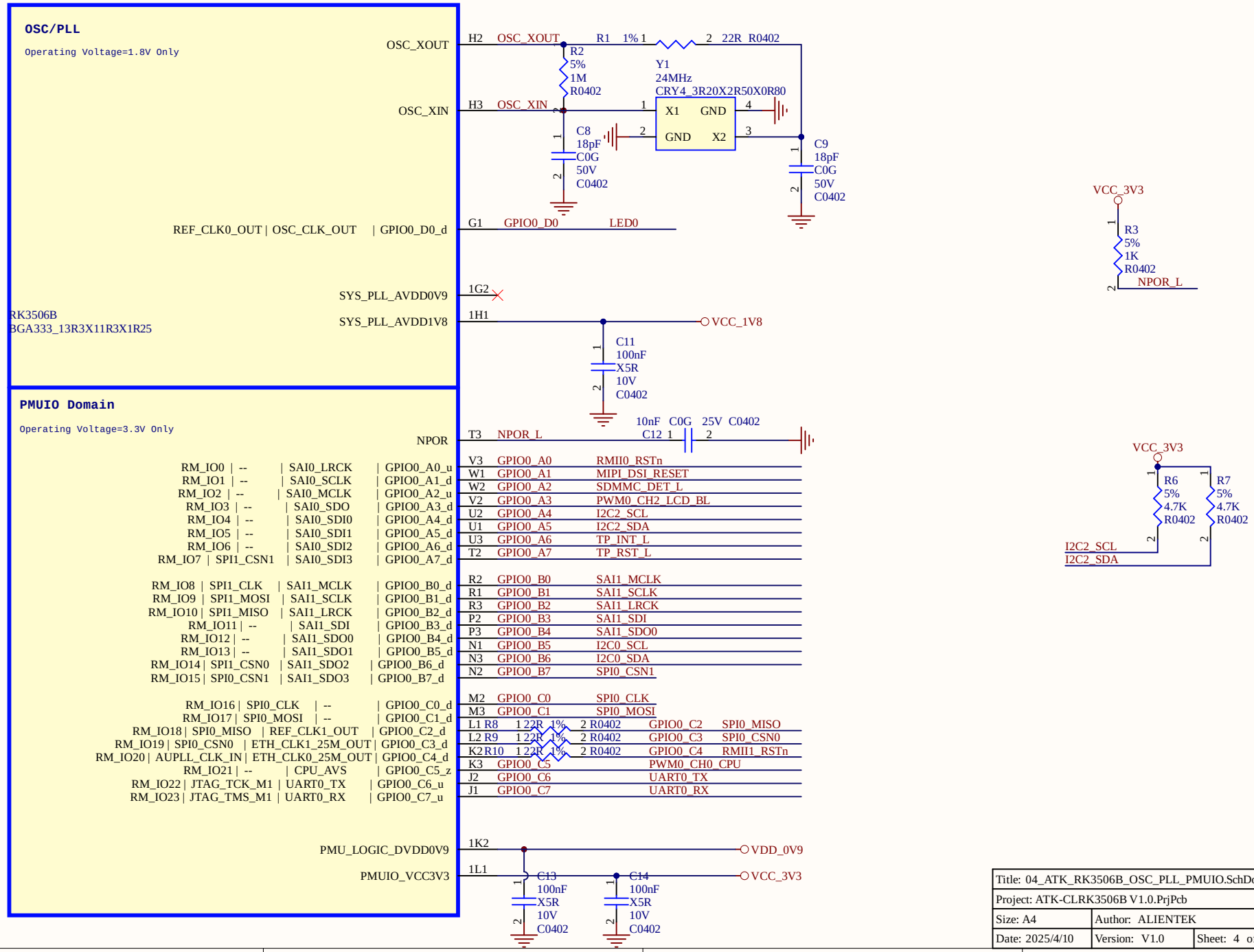
正点原子





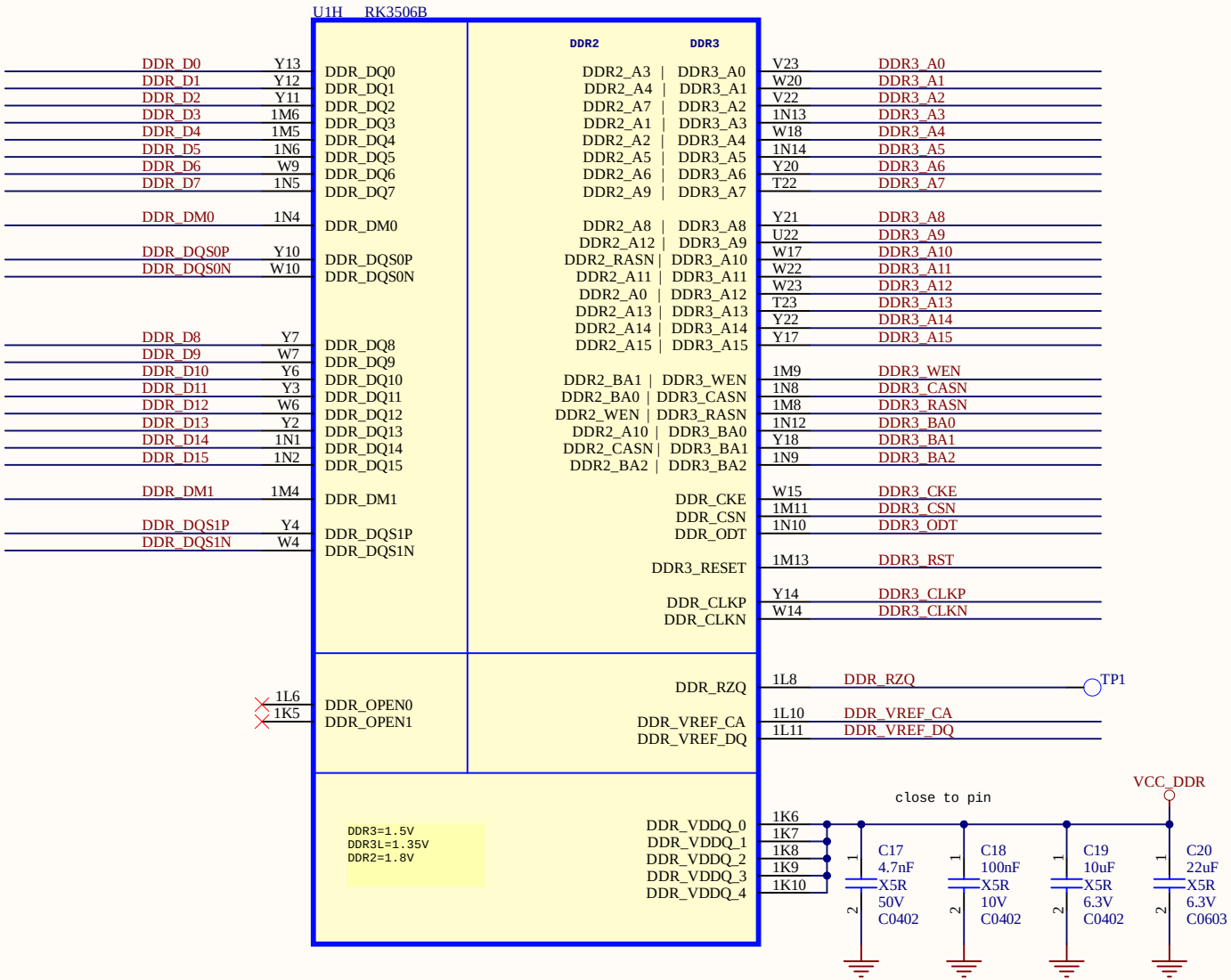
Title: 03_ATK_RK3506B_POWER_GND.SchDoc			
Project: ATK-CLRK3506B V1.0.PrjPcb			
Size: A4	Author: ALIENTEK		
Date: 2025/4/10	Version: V1.0	Sheet: 3 of 14	

正点原子



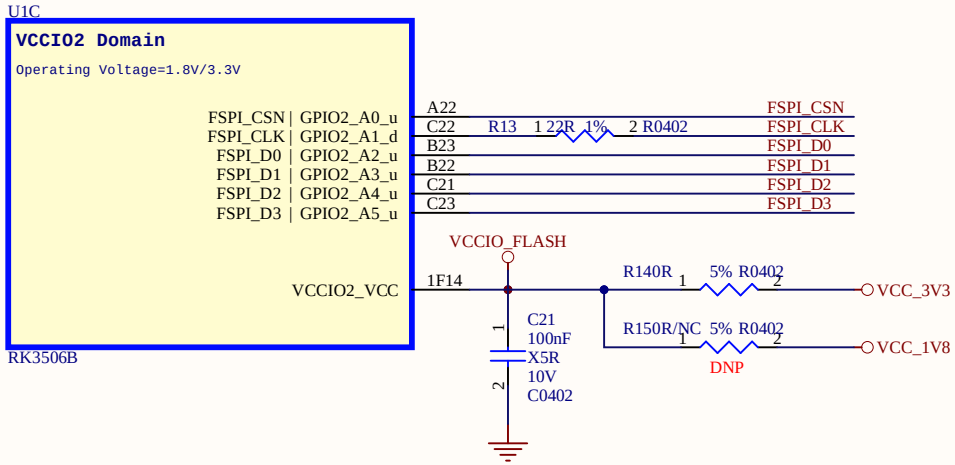
Title: 04_ATK_RK3506B_OSC_PLL_PMUIO.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 4 of 14

正点原子



Title: 05_ATK_RK3506B_DRAM.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 5 of 14

正点原子



Title: 06_ATK_RK3506B_Flash.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 6 of 14

正点原子

U1F

MIPI

Operating Voltage=1.8V

GPO is output only

MIPI_DPHY_DSI_TX_D0N | GPO4_A0_z
MIPI_DPHY_DSI_TX_D0P | GPO4_A1_z
MIPI_DPHY_DSI_TX_D1N | GPO4_A2_z
MIPI_DPHY_DSI_TX_D1P | GPO4_A3_z
MIPI_DPHY_DSI_TX_CLKN | GPO4_A4_z
MIPI_DPHY_DSI_TX_CLKP | GPO4_A5_z

A11 GPIO4_A0 MIPI_DPHY_DSI_TX_D0N
B11 GPIO4_A1 MIPI_DPHY_DSI_TX_D0P
B12 GPIO4_A2 MIPI_DPHY_DSI_TX_D1N
A12 GPIO4_A3 MIPI_DPHY_DSI_TX_D1P
B14 GPIO4_A4 MIPI_DPHY_DSI_TX_CLKN
B13 GPIO4_A5 MIPI_DPHY_DSI_TX_CLKP

MIPI_DPHY_AVDD0V9
MIPI_DPHY_AVDD1V8

VDD_0V9 VCC_1V8

1C7
1B7

C22 100nF X5R 10V C0402
C23 100nF X5R 10V C0402
C24 1uF/NC X5R DNP 10V C0402

USB2.0 OTG0

HS/FS/LS
(Download Port)

USB20_OTG0_DP
USB20_OTG0_DM
USB20_OTG0_ID
USB20_OTG0_VBUSDET

A16 USB20_OTG0_DP
B16 USB20_OTG0_DM
B17 USB20_OTG0_ID
B18 USB20_OTG0_VBUSDET

USB2.0 OTG1

HS/FS/LS

USB20_OTG1_DP
USB20_OTG1_DM

B15 USB20_OTG1_DP
A15 USB20_OTG1_DM

USB2.0 PHY Power

for OTG0/1

USB20_OTG_AVDD0V9
USB20_OTG_AVDD1V8
USB20_OTG_AVDD3V3

VDD_0V9 VCC_1V8 VCC_3V3

1C8
1B9
1B10

C25 100nF X5R 10V C0402
C26 100nF X5R 10V C0402
C27 100nF X5R 10V C0402
C28 1uF/NC X5R DNP 10V C0402

SARADC

Operating Voltage=1.8V

SARADC_IN0 | GPIO4_B0_z
SARADC_IN1 | GPIO4_B1_z
SARADC_IN2 | GPIO4_B2_z
SARADC_IN3 | GPIO4_B3_z

OTP_AVDD1V8 | SARADC_AVDD1V8

A20 SARADC_IN0_BOOT C29 1nF C0G 50V C0402
B20 SARADC_IN1_RECOVER/KEY C30 1nF C0G 50V C0402
A21 GPIO4_B2 SARADC_VIN2_LCD_ID C31 1nF C0G 50V C0402
B21 GPIO4_B3 SARADC_IN3 C32 1nF C0G 50V C0402

1B14 VCC_1V8

C33 100nF X5R 10V C0402

TSADC

TSADC_VCC1V8

1B6 VCC_1V8

C34 100nF X5R 10V C0402

ACODEC

ACODEC_ADC_INP
ACODEC_ADC_INN
ACODEC_ADC_AVDD1V8
ACODEC_ADC_AVDD1V6
ACODEC_ADC_VCM
ACODEC_ADC_AVSS

A19 ACODEC_ADCINP C35 1 2 1uF X5R 10V C0402 ACODEC_ADC_INP
B19 ACODEC_ADCINN C36 1 2 1uF X5R 10V C0402 ACODEC_ADC_INN

VCC_1V8

1B13
1B12 ACODEC_ADC_AVDD1V6
1B11 ACODEC_ADC_VCM
1A13

C37 2.2uF X5R 10V C0402
C38 2.2uF X5R 10V C0402
C39 100nF X5R 10V C0402

RK3506B

SARADC_IN0_BOOT TABLE

Item	Rup	Rdown	ADC	BOOT MODE
LEVEL1	DNP	100K	0	USB(Maskrom mode)/SPI2APB
LEVEL2	100K	12K	110	SPI2APB
LEVEL3	100K	27K	217	FSPI--USB/SPI2APB
LEVEL4	100K	51K	345	
LEVEL5	100K	82K	461	
LEVEL6	100K	120K	558	SDMMC(eMMC/SDCard)--USB/SPI2APB
LEVEL7	100K	200K	682	
LEVEL8	100K	330K	785	
LEVEL9	100K	820K	912	
LEVEL10	10K	DNP	1023	FSPI--SDMMC(eMMC/SDCard)--USB/SPI2APB

VCC_1V8

R21 10K 1% R0402
SARADC_IN0_BOOT
R22 100K/NC 1% DNP R0402

VCC_1V8

R24 10K 1% R0402
SARADC_IN1_RECOVER/KEY

Title: 07_ATK_RK3506B_USB_ACODEC_SARADC_MIPI.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A3	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 7 of 14

正点原子

U1B RK3506B

VCCIO1 Domain

Operating Voltage=1.8V/3.3V

							FLEXBUS1_D0	DSMC_CLKP	VO_LCDC_DEN	GPIO1_A0_d	B10 GPIO1_A0 VO_LCDC_DEN
				DSMC_SLV_INT		DSMC_INT0	FLEXBUS1_D1	DSMC_CLKN	VO_LCDC_VSYNC	GPIO1_A1_d	1A5 GPIO1_A1 R41 22R 2 1% R0402 VO_LCDC_VSYNC
							FLEXBUS1_D2	DSMC_DQS0	VO_LCDC_HSYNC	GPIO1_A2_d	A9 GPIO1_A2 R1D3 22R 2 1% R0402 VO_LCDC_HSYNC
							FLEXBUS1_D3	DSMC_D0	VO_LCDC_CLK	GPIO1_A3_d	B9 GPIO1_A3 R1D4 22R 2 1% R0402 VO_LCDC_CLK
							FLEXBUS1_D4	DSMC_D1	VO_LCDC_D23	GPIO1_A4_d	B8 GPIO1_A4 VO_LCDC_D23
							FLEXBUS1_D5	DSMC_D2	VO_LCDC_D22	GPIO1_A5_d	1A4 GPIO1_A5 VO_LCDC_D22
							FLEXBUS1_D6	DSMC_D3	VO_LCDC_D21	GPIO1_A6_d	A7 GPIO1_A6 VO_LCDC_D21
							FLEXBUS1_D7	DSMC_D4	VO_LCDC_D20	GPIO1_A7_d	B7 GPIO1_A7 VO_LCDC_D20
						FLEXBUS0_CSN_M0	FLEXBUS1_D8	DSMC_D5	VO_LCDC_D19	GPIO1_B0_d	1A3 GPIO1_B0 VO_LCDC_D19
		RM_IO24	UART5_CTSN_M1	FLEXBUS1_CSN_M0			FLEXBUS1_D9	DSMC_CSN1	VO_LCDC_D18	GPIO1_B1_d	1A2 GPIO1_B1 VO_LCDC_D18 I2C1_SCL
	RM_IO25	SAI2_SCLK_M1	FLEXBUS0_CSN_M1	FLEXBUS0_D15		FLEXBUS1_D10	DSMC_INT2	VO_LCDC_D17		GPIO1_B2_d	A5 GPIO1_B2 VO_LCDC_D17 I2C1_SDA
	RM_IO26	SAI2_LRCK_M1	FLEXBUS1_CSN_M1	FLEXBUS0_D14		FLEXBUS1_D11	DSMC_INT3	VO_LCDC_D16		GPIO1_B3_d	B5 GPIO1_B3 VO_LCDC_D16 UART2_TX
			FLEXBUS0_CSN_M2	FLEXBUS0_D13		FLEXBUS1_D12	DSMC_D6	VO_LCDC_D15		GPIO1_B4_d	1A1 GPIO1_B4 VO_LCDC_D15
			FLEXBUS1_CSN_M2	FLEXBUS0_D12		FLEXBUS1_D13	DSMC_D7	VO_LCDC_D14		GPIO1_B5_d	C3 GPIO1_B5 VO_LCDC_D14
		FLEXBUS0_CSN_M3		FLEXBUS0_D11		FLEXBUS1_D14	DSMC_CSN0	VO_LCDC_D13		GPIO1_B6_d	A3 GPIO1_B6 VO_LCDC_D13
		FLEXBUS1_CSN_M3		FLEXBUS0_D10		FLEXBUS1_D15	DSMC_RDYN	VO_LCDC_D12		GPIO1_B7_d	B3 GPIO1_B7 VO_LCDC_D12
DSMC_SLV_CLK		FLEXBUS0_CSN_M4		DSMC_INT1		FLEXBUS1_CLK	DSMC_RESETN	VO_LCDC_D11		GPIO1_C0_d	A2 GPIO1_C0 VO_LCDC_D11
DSMC_SLV_DQS0		SAI2_MCLK_M1	FLEXBUS1_CSN_M4	DSM_AUD_RN_M0		FLEXBUS0_CLK	DSMC_D8	VO_LCDC_D10		GPIO1_C1_d	B2 GPIO1_C1 VO_LCDC_D10
DSMC_SLV_D0	RM_IO27	SAI2_SDI_M1	FLEXBUS0_CSN_M5	DSM_AUD_RP_M0		FLEXBUS0_D9	DSMC_D9	VO_LCDC_D9		GPIO1_C2_d	B1 GPIO1_C2 VO_LCDC_D9 UART2_RX
DSMC_SLV_D1	RM_IO28	SAI2_SDO_M1	FLEXBUS1_CSN_M5			FLEXBUS0_D8	DSMC_D10	VO_LCDC_D8		GPIO1_C3_d	C2 GPIO1_C3 VO_LCDC_D8
DSMC_SLV_D2						FLEXBUS0_D7	DSMC_D11	VO_LCDC_D7		GPIO1_C4_d	C1 GPIO1_C4 VO_LCDC_D7
DSMC_SLV_D3						FLEXBUS0_D6	DSMC_D12	VO_LCDC_D6		GPIO1_C5_d	D3 GPIO1_C5 VO_LCDC_D6
DSMC_SLV_D4						FLEXBUS0_D5	DSMC_D13	VO_LCDC_D5		GPIO1_C6_d	E3 GPIO1_C6 VO_LCDC_D5
DSMC_SLV_D5						FLEXBUS0_D4	DSMC_D14	VO_LCDC_D4		GPIO1_C7_d	E2 GPIO1_C7 VO_LCDC_D4
DSMC_SLV_D6				DSM_AUD_LN_M0		FLEXBUS0_D3	DSMC_D15	VO_LCDC_D3		GPIO1_D0_d	E1 GPIO1_D0 VO_LCDC_D3
DSMC_SLV_D7	RM_IO29	UART5_RTSN_M1		DSM_AUD_LP_M0		FLEXBUS0_D2	DSMC_DQS1	VO_LCDC_D2		GPIO1_D1_d	F3 GPIO1_D1 VO_LCDC_D2
DSMC_SLV_CSN0	RM_IO30	UART5_TX_M1				FLEXBUS0_D1	DSMC_CSN2	VO_LCDC_D1		GPIO1_D2_d	F2 GPIO1_D2 VO_LCDC_D1
DSMC_SLV_RDYN	RM_IO31	UART5_RX_M1				FLEXBUS0_D0	DSMC_CSN3	VO_LCDC_D0		GPIO1_D3_d	G3 GPIO1_D3 VO_LCDC_D0

VCCIO1_VCC_0
VCCIO1_VCC_1

1E3
1F3

C40 100nF X5R 10V C0402 2

C41 100nF X5R 10V C0402 2

VCC_3V3

Mode	RGB					BT1120	BT656	MCU				
	24bit	18bit	16bit	8bit	6bit	16bit	8bit	24bit	18bit	16bit	8bit	6bit
LCDC_DEN	DEN	DEN	DEN	DEN	DEN	--	--	RDN	RDN	RDN	RDN	RDN
LCDC_VSYNC	VSYNC	VSYNC	VSYNC	VSYNC	VSYNC	--	--	CSN	CSN	CSN	CSN	CSN
LCDC_HSYNC	HSYNC	HSYNC	HSYNC	HSYNC	HSYNC	--	--	WRN	WRN	WRN	WRN	WRN
LCDC_CLK	CLK	CLK	CLK	CLK	CLK	CLK	CLK	RS	RS	RS	RS	RS
LCDC_D23	D23	D17	D15	D7_m1	D5_m1	D15	D7_m1	D23	D17	D15	D7_m1	D5_m1
LCDC_D22	D22	D16	D14	D6_m1	D4_m1	D14	D6_m1	D22	D16	D14	D6_m1	D4_m1
LCDC_D21	D21	D15	D13	D5_m1	D3_m1	D13	D5_m1	D21	D15	D13	D5_m1	D3_m1
LCDC_D20	D20	D14	D12	D4_m1	D2_m1	D12	D4_m1	D20	D14	D12	D4_m1	D2_m1
LCDC_D19	D19	D13	D11	D3_m1	D1_m1	D11	D3_m1	D19	D13	D11	D3_m1	D1_m1
LCDC_D18	D18	D12	--	--	--	--	--	D18	D12	--	--	--
LCDC_D17	D17	--	--	--	--	--	--	D17	--	--	--	--
LCDC_D16	D16	--	--	--	--	--	--	D16	--	--	--	--
LCDC_D15	D15	D11	D10	D2_m1	D0_m1	D10	D2_m1	D15	D11	D10	D2_m1	D0_m1
LCDC_D14	D14	D10	D9	D1_m1	--	D9	D1_m1	D14	D10	D9	D1_m1	--
LCDC_D13	D13	D9	D8	D0_m1	--	D8	D0_m1	D13	D9	D8	D0_m1	--
LCDC_D12	D12	D8	D7	D7_m0	D5_m0	D7	D7_m0	D12	D8	D7	D7_m0	D5_m0
LCDC_D11	D11	D7	D6	D6_m0	D4_m0	D6	D6_m0	D11	D7	D6	D6_m0	D4_m0
LCDC_D10	D10	D6	D5	D5_m0	D3_m0	D5	D5_m0	D10	D6	D5	D5_m0	D3_m0
LCDC_D9	D9	--	--	--	--	--	--	D9	--	--	--	--
LCDC_D8	D8	--	--	--	--	--	--	D8	--	--	--	--
LCDC_D7	D7	D5	D4	D4_m0	D2_m0	D4	D4_m0	D7	D5	D4	D4_m0	D2_m0
LCDC_D6	D6	D4	D3	D3_m0	D1_m0	D3	D3_m0	D6	D4	D3	D3_m0	D1_m0
LCDC_D5	D5	D3	D2	D2_m0	D0_m0	D2	D2_m0	D5	D3	D2	D2_m0	D0_m0
LCDC_D4	D4	D2	D1	D1_m0	--	D1	D1_m0	D4	D2	D1	D1_m0	--
LCDC_D3	D3	D1	D0	D0_m0	--	D0	D0_m0	D3	D1	D0	D0_m0	--
LCDC_D2	D2	D0	--	--	--	--	--	D2	D0	--	--	--
LCDC_D1	D1	--	--	--	--	--	--	D1	--	--	--	--
LCDC_D0	D0	--	--	--	--	--	--	D0	--	--	--	--

Title: 08_ATK_RK3506B_VO_GPIO.SchDoc			
Project: ATK-CLRK3506B V1.0.PrjPcb			
Size: A3	Author: ALIENTEK		
Date: 2025/4/10	Version: V1.0	Sheet: 8 of 14	

正点原子

U1D

VCCIO3 Domain

Operating Voltage=1.8V/3.3V

SPI2 only support slave

--	SPI2_CLK	ETH_RMII0_RXD0
--	SPI2_CSN	ETH_RMII0_RXD1
--	SPI2_MOSI	ETH_RMII0_CLK
--	SPI2_MISO	ETH_RMII0_TXD0
SAI3_SCLK	DSM_AUD_RN_M1	ETH_RMII0_TXD1
SAI3_LRCK	DSM_AUD_RP_M1	ETH_RMII0_TXEN
SAI3_SDI	DSM_AUD_LN_M1	ETH_RMII0_MDC
SAI3_SDO	DSM_AUD_LP_M1	ETH_RMII0_MDIO
SAI3_MCLK	--	ETH_RMII0_RXDVCRS

GPIO2_B0_d	D21		GPIO2_B0	RMII0_RXD0
GPIO2_B1_d	D23		GPIO2_B1	RMII0_RXD1
GPIO2_B2_d	E21	R28 22R1 2 1% R0402	GPIO2_B2	RMII0_CLK
GPIO2_B3_d	E22	R29 22R1 2 1% R0402	GPIO2_B3	RMII0_TXD0
GPIO2_B4_d	F21	R30 22R1 2 1% R0402	GPIO2_B4	RMII0_TXD1
GPIO2_B5_d	F23	R31 22R1 2 1% R0402	GPIO2_B5	RMII0_TXEN
GPIO2_B6_d	F22		GPIO2_B6	RMII0_MDC
GPIO2_B7_d	G21		GPIO2_B7	RMII0_MDIO
GPIO2_C0_d	G22		GPIO2_C0	RMII0_RXDV_CRS

VCCIO3_VCC — 1G14 — C42 100nF X5R 10V C0402 — VCC_3V3

RK3506B

U1E

VCCIO4 Domain

Operating Voltage=1.8V/3.3V

--	SDMMC_CLK	GPIO3_A0_d
--	SDMMC_CMD	GPIO3_A1_d
--	SDMMC_D0	GPIO3_A2_d
TEST_CLK_OUT	SDMMC_D1	GPIO3_A3_d
JTAG_TCK_M0	SDMMC_D2	GPIO3_A4_d
JTAG_TMS_M0	SDMMC_D3	GPIO3_A5_d
ETH_RMII1_RXD0	SAI2_SDI_M0	GPIO3_A6_d
ETH_RMII1_RXD1	SAI2_SCLK_M0	GPIO3_A7_d
ETH_RMII1_CLK	SAI2_SDO_M0	GPIO3_B0_d
ETH_RMII1_TXD0	SAI2_LRCK_M0	GPIO3_B1_d
ETH_RMII1_TXD1	UART5_CTSN_M0	GPIO3_B2_d
ETH_RMII1_TXEN	UART5_RX_M0	GPIO3_B3_d
ETH_RMII1_MDC	UART5_TX_M0	GPIO3_B4_d
ETH_RMII1_MDIO	UART5_RTSN_M0	GPIO3_B5_d
ETH_RMII1_RXDVCRS	SAI2_MCLK_M0	GPIO3_B6_d

GPIO3_A0_d	K21		GPIO3_A0	SDMMC_CLK
GPIO3_A1_d	J21		GPIO3_A1	SDMMC_CMD
GPIO3_A2_d	K23		GPIO3_A2	SDMMC_D0
GPIO3_A3_d	K22		GPIO3_A3	SDMMC_D1
GPIO3_A4_d	H23		GPIO3_A4	SDMMC_D2
GPIO3_A5_d	H22		GPIO3_A5	SDMMC_D3
GPIO3_A6_d	L22		GPIO3_A6	RMII1_RXD0
GPIO3_A7_d	M22		GPIO3_A7	RMII1_RXD1
GPIO3_B0_d	M23	R33 22R1 2 1% R0402	GPIO3_B0	RMII1_CLK
GPIO3_B1_d	M21	R34 22R1 2 1% R0402	GPIO3_B1	RMII1_TXD0
GPIO3_B2_d	N21	R35 22R1 2 1% R0402	GPIO3_B2	RMII1_TXD1
GPIO3_B3_d	P22	R36 22R1 2 1% R0402	GPIO3_B3	RMII1_TXEN
GPIO3_B4_d	P23		GPIO3_B4	RMII1_MDC
GPIO3_B5_d	P21		GPIO3_B5	RMII1_MDIO
GPIO3_B6_d	R22		GPIO3_B6	RMII1_RXDV_CRS

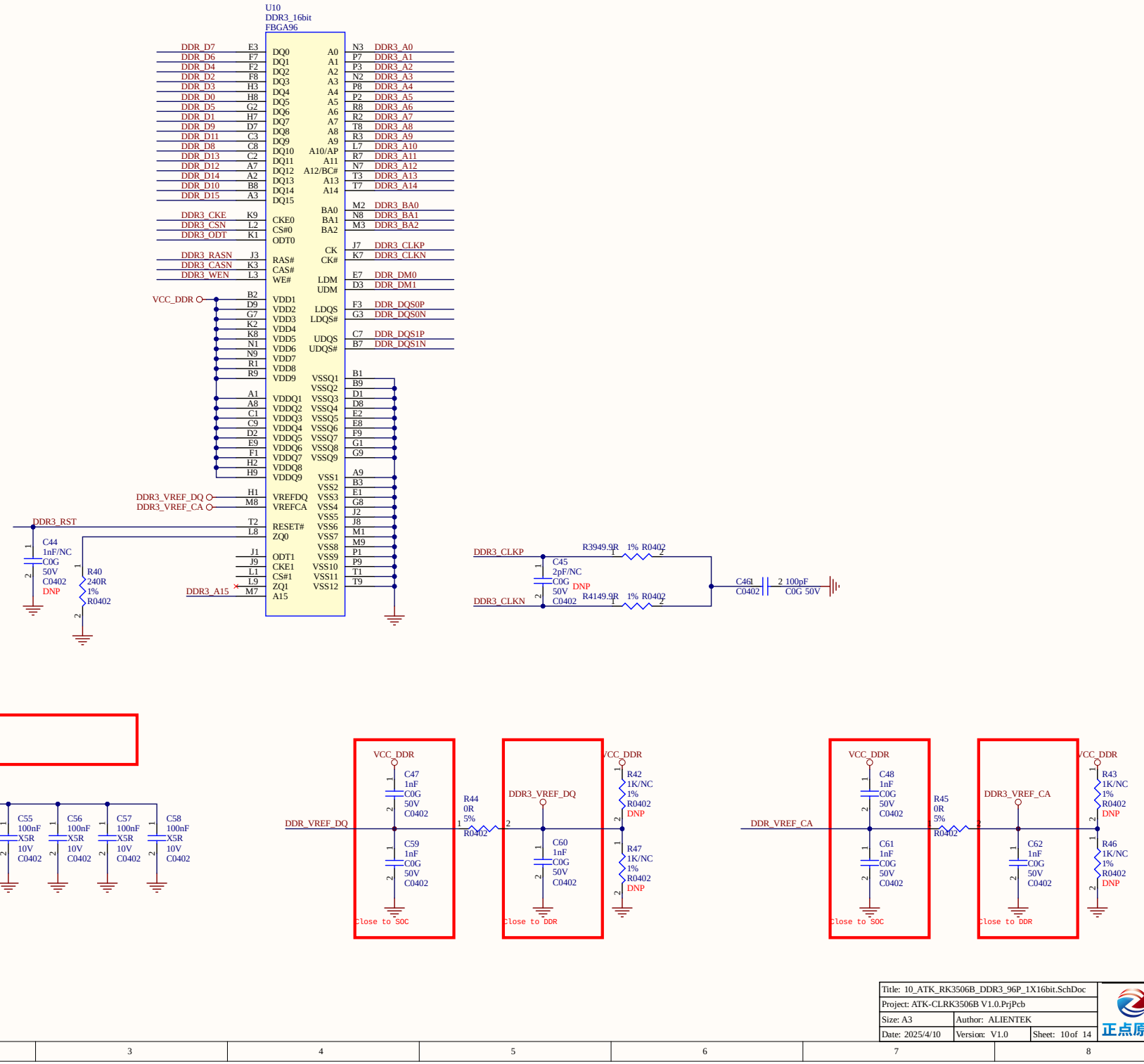
VCCIO4_VCC — 1H14 — C43 100nF X5R 10V C0402 — R370R 1 5% R0402 2 — VCC_3V3

R380R/NC 1 5% R0402 2 DNP — VCC_1V8

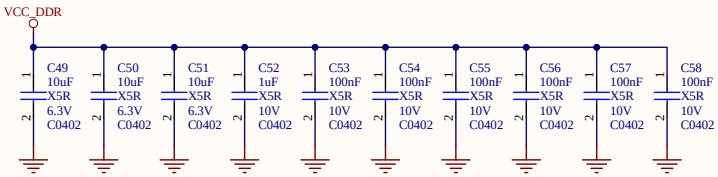
RK3506B

Title: 09_ATK_RK3506B_RMII_SDMMC_GPIO.SchDoc			
Project: ATK-CLRK3506B V1.0.PrjPcb			
Size: A4	Author: ALIENTEK		
Date: 2025/4/10	Version: V1.0	Sheet: 9 of 14	

正点原子



Note:All the Power filter capacitors should be placed close to the power pins of DDR

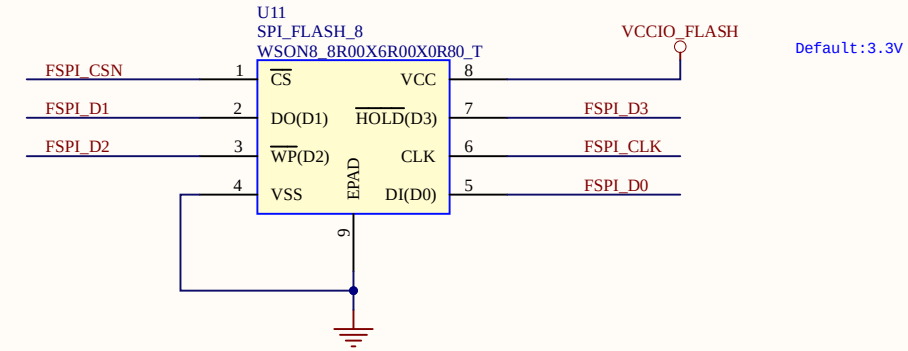


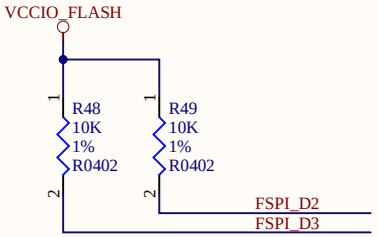
Title: 10_ATK_RK3506B_DDR3_96P_1X16bit.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A3	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 10of 14

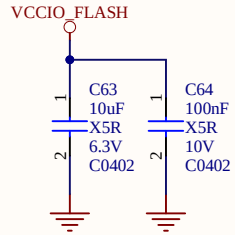
正点原子

SPI Flash

NOTE:
Refer to the latest AVL for parts selection.



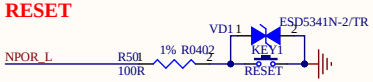




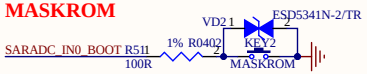
Title: 11_ATK_RK3506B_SPI_NAND_Flash.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 11 of 14

正点原子

RESET

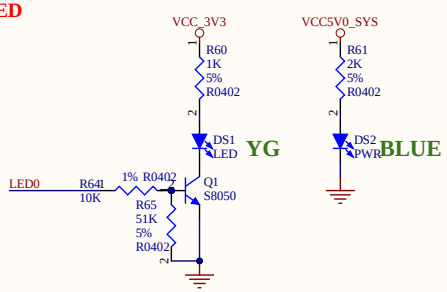


MASKROM

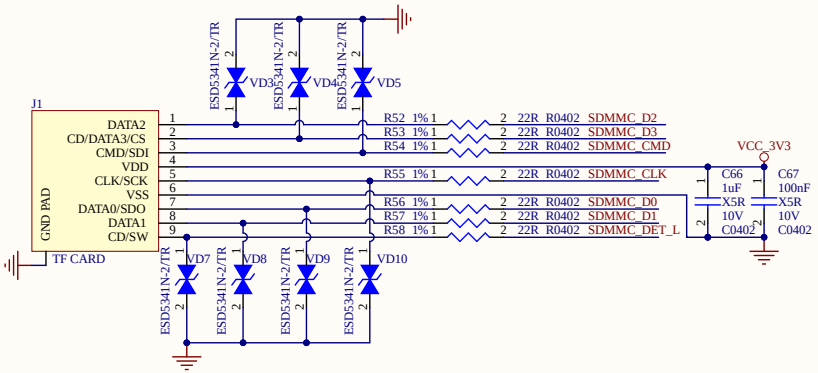




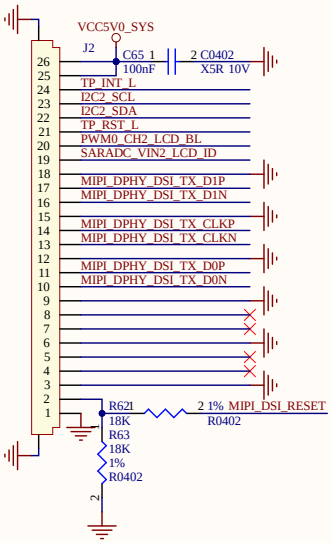
LED



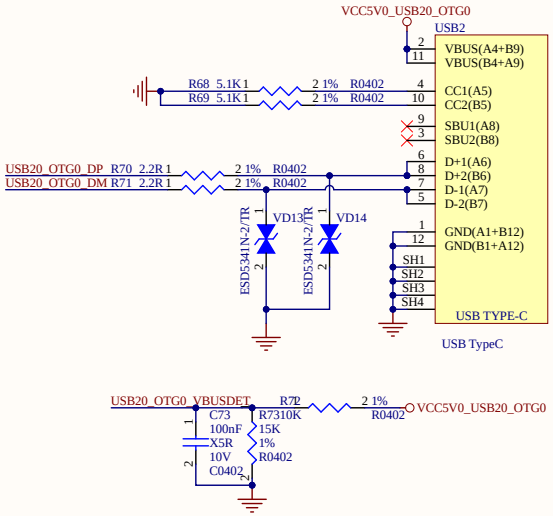
TF Card



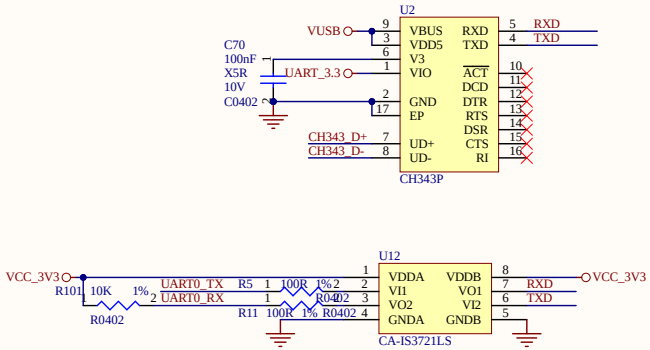
MIPI DSI

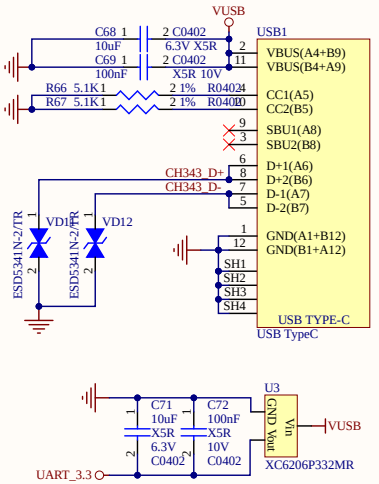


USB



USART(TYPEC)

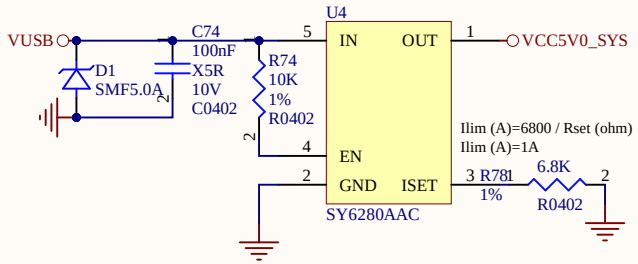




Title: 12_ATK_RK3506B_DEVICE.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 12of 14

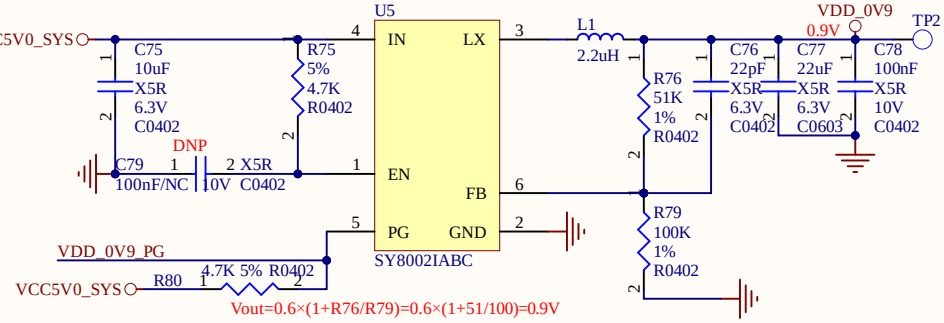
正点原子

USB POWER



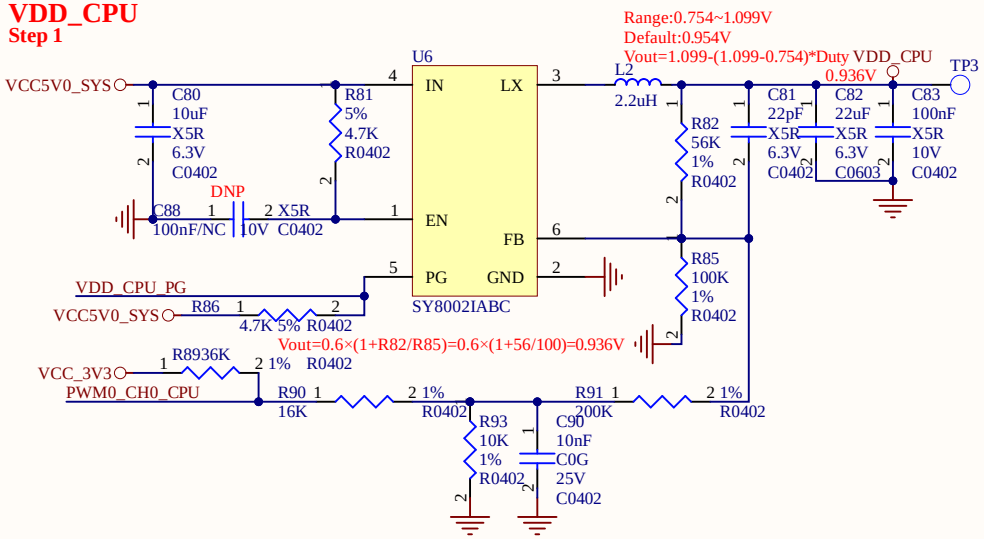
VDD_0V9

Step 1



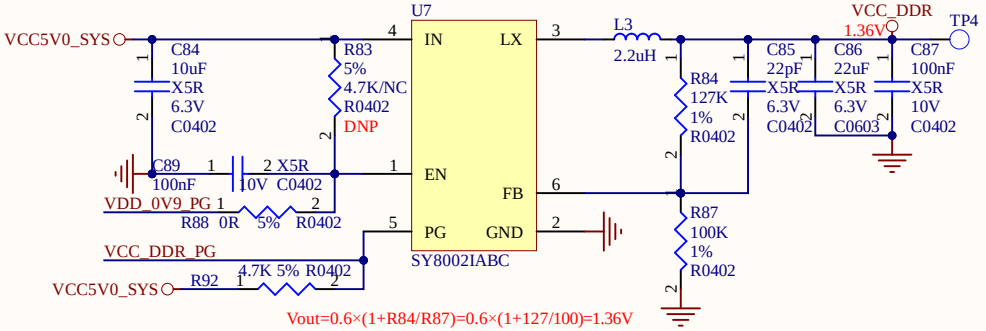
VDD_CPU

Step 1



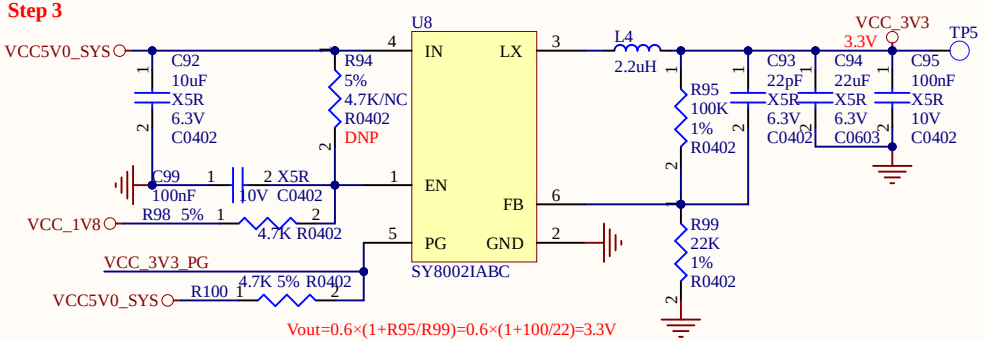
VDD_DDR

Step 2



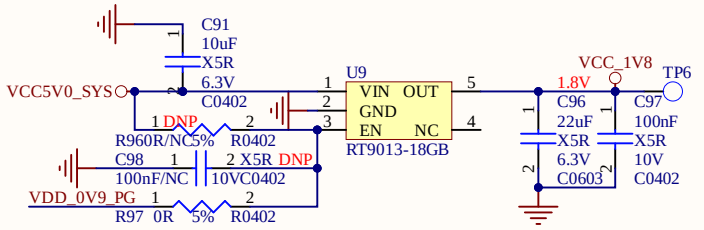
VCC_3V3

Step 3



VCC_1V8

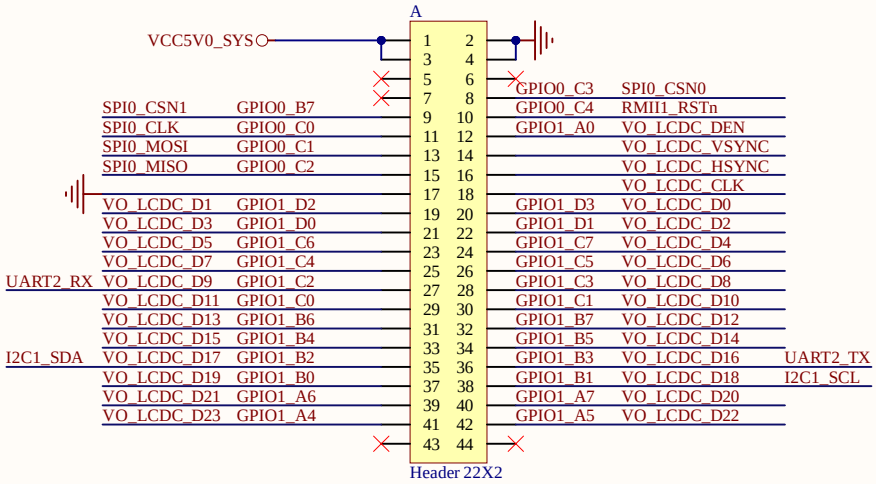
Step 2a

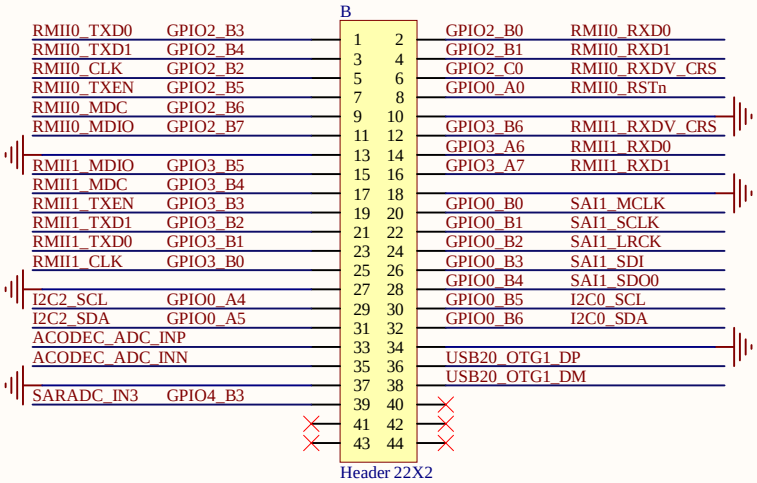


Title: 13_ATK_RK3506B_POWER.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A3	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 13 of 14

正点原子

PINOUT





Title: 14_ATK_RK3506B_PINOUT.SchDoc		
Project: ATK-CLRK3506B V1.0.PrjPcb		
Size: A4	Author: ALIENTEK	
Date: 2025/4/10	Version: V1.0	Sheet: 14 of 14

正点原子